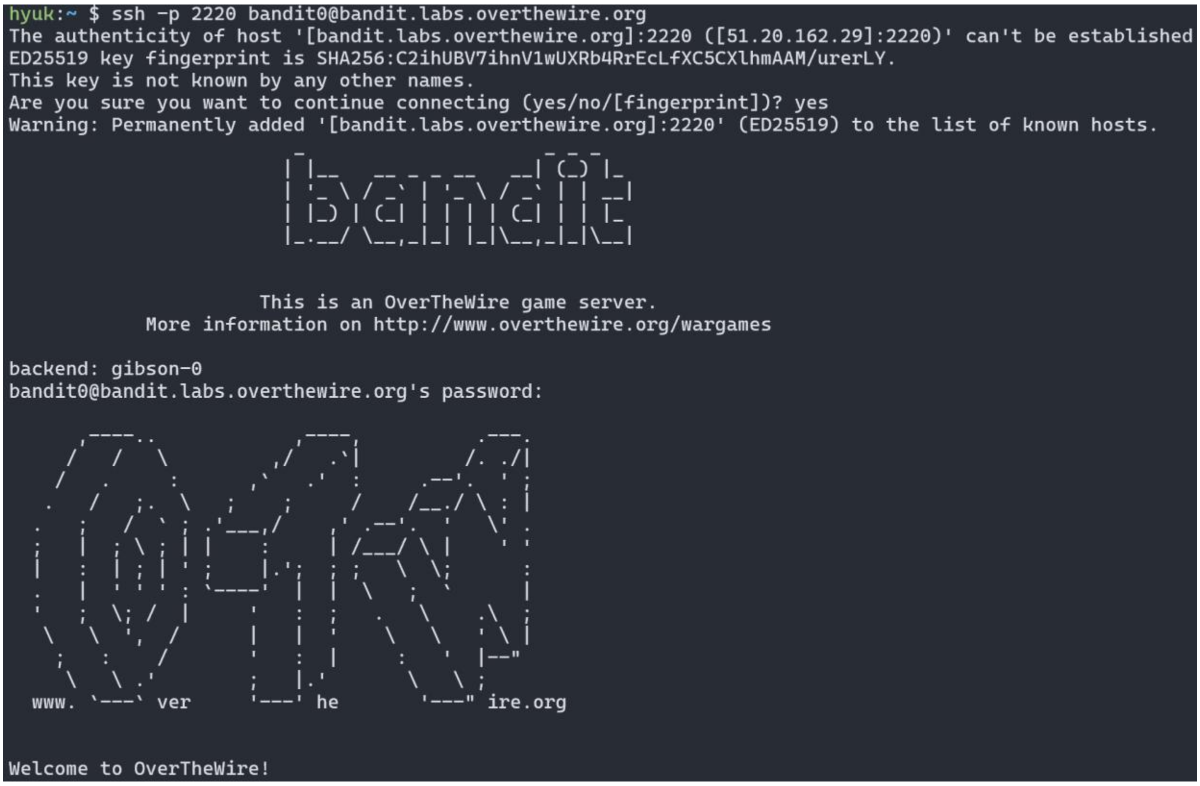
Bandit

Level 0

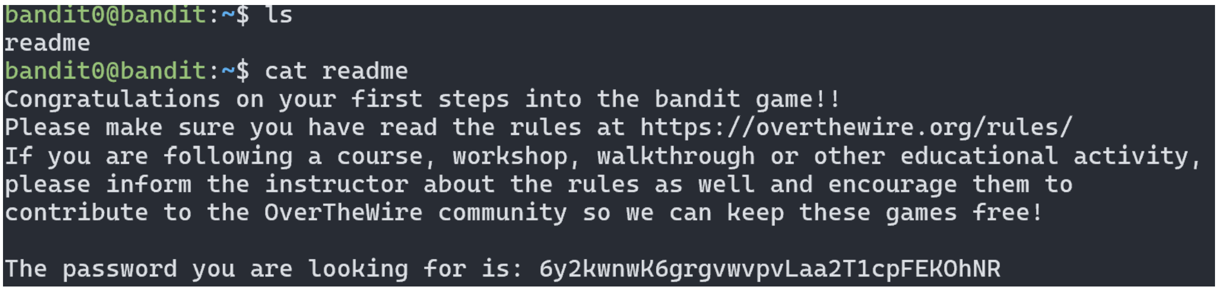
solution: ssh -p 2220 bandit0@bandit.labs.overthewire.org



Level 0 -> Level 1

solution: cat readme

password: 6y2kwnwK6grgvwvpvLaa2T1cpFEKOhNR



Level 1 -> Level 2

solution: cat ./- or cat <-

- 이름의 파일은 ./이나 <을 이용해서 읽음.

password: PK8fYLZg2hnHSz83plBL1iEPKdD3QToB

```
bandit1@bandit:~$ cat ./-
PK8fYLZg2hnHSz83plBL1iEPKdD3QToB
bandit1@bandit:~$ cat <-
PK8fYLZg2hnHSz83plBL1iEPKdD3QToB
```

Level 2 -> Level 3

solution: cat ./”—spaces in this filename—” or cat <”—spaces in this filename—"

파일명에 띄어쓰기가 있으면 “”로 감싸서 읽음.

password: 7ZZ2LFrykP2zEyvBl4m3clcL7tGYJPME

```
bandit2@bandit:~$ cat ./"--spaces in this filename--"
7ZZ2LFrykP2zEyvBl4m3clcL7tGYJPME
bandit2@bandit:~$ cat <"--spaces in this filename--"
7ZZ2LFrykP2zEyvBl4m3clcL7tGYJPME
```

Level 3 -> Level 4

solution: ls -a -> cat …Hiding-From-You

ls의 -a 옵션은 숨겨진 파일까지 드러냄.

password: xzTXq1rDJQVVAzdv5cHq1TQytTWufAMq

```
bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -a
.  ..  ...Hiding-From-You
bandit3@bandit:~/inhere$ cat ...Hiding-From-You
xzTXq1rDJQVVAzdv5cHq1TQytTWufAMq
```

Level 4 -> Level 5

solution: file ./-file* -> cat ./-file07

file [파일명]은 파일의 종류를 확인.

password: 6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG

```
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls
-file00  -file01  -file02  -file03  -file04  -file05  -file06  -file07  -file08  -file09
bandit4@bandit:~/inhere$ cat ./-file00
Y1[�b1���� ��2�D¬1K0
                    3'���bandit4@bandit:~/inhere$
bandit4@bandit:~/inhere$ file ./-file*
./-file00: data
./-file01: data
./-file02: data
./-file03: data
./-file04: data
./-file05: data
./-file06: OpenPGP Public Key
./-file07: ASCII text
./-file08: data
./-file09: Motorola S-Record; binary data in text format
bandit4@bandit:~/inhere$ cat ./-file07
6C7h9GD8M6ai5nr7wo1RonrzFjj9yIrG
```

Level 5 -> Level 6

solution: find -size 1033c -> cat maybehere07/.file2

find는 재귀적으로 모든 하위 디렉터리까지 내려가면서 탐색. size 옵션을 사용하면 특정 크기의 파일을 찾을 수 있음.

password: pXa26xhMWaC2SvDotA4r9EgZkulOeSBW

```
bandit5@bandit:~$ ls
inhere
bandit5@bandit:~$ cd inhere
bandit5@bandit:~/inhere$ ls
maybehere00  maybehere03  maybehere06  maybehere09  maybehere12  maybehere15  maybehere18
maybehere01  maybehere04  maybehere07  maybehere10  maybehere13  maybehere16  maybehere19
maybehere02  maybehere05  maybehere08  maybehere11  maybehere14  maybehere17
bandit5@bandit:~/inhere$ find -size 1033c
./maybehere07/.file2
bandit5@bandit:~/inhere$ cat maybehere07/.file2
pXa26xhMWaC2SvDotA4r9EgZkulOeSBW
```

Level 6 -> Level 7

solution: find / -user bandit7 -group bandit6 -size 33c

find 명령어의 시작 경로를 지정해 줄 수 있음. 리다이렉션을 이용해서 에러가 발생하는 파일을 제외할 수 있음.

password: Bmnnvf82KzQlfxgAI2d1zYbr1u9pr3E3

```
bandit6@bandit:~$ ls
bandit6@bandit:~$ find / -user bandit7 -group bandit6 -size 33c
find: '/snap': Permission denied
find: '/lost+found': Permission denied
find: '/manpage/manpage3-pw': Permission denied
find: '/drifter/drifter14_src/axTLS': Permission denied
find: '/run/pam_pidns': Permission denied
find: '/run/udisks2': Permission denied
find: '/run/chrony': Permission denied
find: '/run/user/11014': Permission denied
find: '/run/user/14001': Permission denied
find: '/run/user/11021': Permission denied
find: '/run/user/11022': Permission denied
find: '/run/user/11026': Permission denied
find: '/run/user/11009': Permission denied
find: '/run/user/11010': Permission denied
find: '/run/user/14000': Permission denied
find: '/run/user/11024': Permission denied
find: '/run/user/12002': Permission denied
find: '/run/user/11015': Permission denied
find: '/run/user/11023': Permission denied
find: '/run/user/11003': Permission denied
find: '/run/user/11004': Permission denied
```

```
bandit6@bandit:~$ find / -user bandit7 -group bandit6 -size 33c 2>/dev/null
/var/lib/dpkg/info/bandit7.password
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
Bmnnvf82KzQlfxgAI2d1zYbr1u9pr3E3
```

Level 7 -> Level 8

solution: grep 'millionth' data.txt

password: VR1ljMayciFxbnUokuQmJFw6QC9VKtub

grep '문자열' [파일명]은 파일 내에서 특정 문자열을 찾음.

```
bandit7@bandit:~$ ls
data.txt
bandit7@bandit:~$ grep 'millionth' data.txt
millionth       VR1ljMayciFxbnUokuQmJFw6QC9VKtub
```

Level 8 -> Level 9

solution: sort data.txt | uniq -u

uniq의 -u 옵션은 한 번만 등장한 행을 화면에 보여줌. 연속된 행끼리만 비교할 수 있기 때문에 sort로 정렬을 한 결과를 파이프를 통해 입력으로 보내줘야 함.

password: EjmOSvuAu7sGAHqHVcBDPirRe9T03kxl

```
bandit8@bandit:~$ ls
data.txt
bandit8@bandit:~$ sort data.txt | uniq -u
EjmOSvuAu7sGAHqHVcBDPirRe9T03kxl
```

Level 9 -> Level 10

solution: strings data.txt | grep '='

strings는 사람이 읽을 수 있는 문자열만 출력함.

password: B0s2khmbT9u0geKuOoVGW3JZKhndE3BG

```
bandit9@bandit:~$ strings data.txt | grep '='
=KGEn
cL0========== the
=<P&
========== password
bU=\
a7=P
>========== is
wbp=
=lR(
a=-io
R========== B0s2khmbT9u0geKuOoVGW3JZKhndE3BG
)=Lc
x=E$
```

Level 10 -> Level 11

solution: cat data.txt | base64 -d

base64의 -d 옵션은 base64로 인코딩된 값을 디코딩함.

password: pYfOY6HwUsDj5rL9UvyhU7MCmv8vN5Ro

```
bandit10@bandit:~$ ls
data.txt
bandit10@bandit:~$ cat data.txt
VGhlIHBhc3N3b3JkIGlzIHBZZk9ZNkh3VXNEajVyTDlVdnloVTdNQ21OHZONVJvCg==
bandit10@bandit:~$ cat data.txt | base64 -d
The password is pYfOY6HwUsDj5rL9UvyhU7MCmv8vN5Ro
```

Level 11 -> Level 12

solution: cat data.txt | tr ‘a-zA-Z’ ‘n-za-mN-ZA-M’

password: GROozWPO8QyN0mGrjUkID0WCYkZiQxrN

```
bandit11@bandit:~$ cat data.txt | tr 'a-zA-Z' 'n-za-mN-ZA-M'
The password is GROozWPO8QyN0mGrjUkID0WCYkZiQxrN
```

Level 12 -> Level 13

solution:

gzip, bzip2는 압축 도구, tar은 파일을 여러 개로 묶는 도구. data.txt는 hexdump된 파일이므로 xxd -r을 이용해 바이너리 파일로 변경 후 파일 종류를 확인하니 gzip으로 압축된 파일임을 확인. 홈 디렉터리에서는 파일을 생성할 수 없어서 mktemp -d로 임시 디렉터리를 생성하여 임시 디렉터리에서 해결. file 명령어로 파일 종류를 확인인 후 mv 명령어로 파일 이름을 바꿔서 압축 해제.

gzip과 bzip2는 -d 옵션을 사용하여 압축을 해제할 수 있음. tar의 x옵션은 아카이브 해제, f 옵션은 파일명 지정, v 옵션은 처리 과정 나열.

password: qQYQiHOBPR8zR61qxYqX45quvihF2uzk

```
bandit12@bandit:~$ cat data.txt
00000000: 1f8b 0808 b2f0 3b6a 0203 6461 7461 322e  ......;j..data2.
00000010: 6269 6e00 0142 02bd fd42 5a68 3931 4159  bin..B...BZh91AY
00000020: 2653 59dc 0966 8300 001a ffff dff5 c5fe  &SY..f..........
00000030: b8ef a7be bddb f8a7 febb ffc9 bfbf 9fbf  ................
00000040: b77b bfff fbd9 7ffe 5fef efcf b001 3b19  .{......_.....;.
00000050: 9206 87a9 a068 0340 3400 341a 1a1a 0340  .....h.@4.4....@
00000060: 1a68 068d 1a00 0646 8000 0610 0d00 069a  .h.....F........
00000070: 0640 683c a0d0 3d43 4d3d 4f51 b420 0680  .@h<..=CM=OQ. ..
00000080: d034 0000 0079 41a3 40d1 ea00 0f48 000d  .4...yA.@....H..
```

```
bandit12@bandit:~$ xxd -r data.txt > bin
-bash: bin: Permission denied
```

```
bandit12@bandit:~$ mktemp -d
/tmp/tmp.q1qOFh51KL
bandit12@bandit:~$ cp data.txt /tmp/tmp.q1qOFh51KL
bandit12@bandit:~$ cd /tmp/tmp.q1qOFh51KL
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ ls
data.txt
```

```
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ xxd -r data.txt > bin
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file bin
bin: gzip compressed data, was "data2.bin", last modified: Wed Jun 24 14:58:58 2026, max compression, from Unix, origina
l size modulo 2^32 578
```

```
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ mv bin bin.gz
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ gzip bin.gz -d
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ ls
bin  data.txt
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file bin
bin: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ mv bin bin.bz2
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ bzip2 bin.bz2 -d
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ ls
bin  data.txt
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file bin
bin: gzip compressed data, was "data4.bin", last modified: Wed Jun 24 14:58:58 2026, max compression, from Unix, origina
l size modulo 2^32 20480
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ mv bin bin.gz
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ gzip bin.gz -d
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ ls
bin  data.txt
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file bin
bin: POSIX tar archive (GNU)
```

```
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ tar -xvf bin
data5.bin
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file data5.bin
data5.bin: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ tar -xvf data5.bin
data6.bin
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file data6.bin
data6.bin: bzip2 compressed data, block size = 900k
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ mv data6.bin data6.bz2
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ bzip2 data6.bz2 -d
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ ls
bin  data.txt  data5.bin  data6
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file data6
data6: POSIX tar archive (GNU)
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ tar -xvf data6
data8.bin
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file data8.bin
data8.bin: gzip compressed data, was "data9.bin", last modified: Wed Jun 24 14:58:58 2026, max compression, from Unix, o
riginal size modulo 2^32 49
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ mv data8.bin data8.gz
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ gzip data8.gz -d
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ ls
bin  data.txt  data5.bin  data6  data8
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ file data8
data8: ASCII text
bandit12@bandit:/tmp/tmp.q1qOFh51KL$ cat data8
The password is qQYQiHOBPR8zR61qxYqX45quvihF2uzk
```

Level 13 -> Level 14

solution: scp -P 2220 bandit13@bandit.labs.overthewire.org:sshkey.private -> chmod 600 sshkey.private -> ssh -i sshkey.private -p 2220 bandit14@bandit.labs.overthewire.org

scp는 원격 서버에서 파일을 로컬로 복사할 수 있음. chmod는 파일의 권한을 변경. 소유자, 그룹, 그 외 사용자 순서. 1은 실행, 2는 쓰기, 4는 읽기 권한이며 이를 조합해서 읽기와 쓰기 권한 = 6 같은 조합을 만듦.

password: aaWecNkG4FhxJQxz07uiwzVP6bJiYS65

```
hyuk:~ $ scp -P 2220 bandit13@bandit.labs.overthewire.org:sshkey.private .

                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit13@bandit.labs.overthewire.org's password:
sshkey.private                                              100% 2602     3.1KB/s   00:00
```

```
hyuk:~ $ chmod 600 sshkey.private
hyuk:~ $ ssh -i sshkey.private -p 2220 bandit14@bandit.labs.overthewire.org

                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0

  www.       ver          he          ire.org

Welcome to OverTheWire!
```

```
bandit14@bandit:/etc/bandit_pass$ cat bandit14
aaWecNkG4FhxJQxz07uiwzVP6bJiYS65
```

Level 14 -> Level 15

solution: nc localhost 30000

nc는 TCP 또는 UDP 포트를 통한 네트워크 연결, 데이터 송수신, 포트 오픈 여부 확인에 사용되는 다목적 네트워크 유틸리티.

password: pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7

```
bandit14@bandit:/etc/bandit_pass$ cat bandit14
aaWecNkG4FhxJQxz07uiwzVP6bJiYS65
bandit14@bandit:/etc/bandit_pass$ nc localhost 30000
aaWecNkG4FhxJQxz07uiwzVP6bJiYS65
Correct!
pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7
```

Level 15 -> Level 16

solution: openssl s_client -connect localhost:30001 or ncat –ssl localhost 30001

openssl s_client는 원격 서버의 SSL/TLS 연결을 테스트하고 서버가 제공하는 SSL 인증서 정보, 지원하는 암호화 방식 등을 확인. 보안 암호화(SSL/TLS)가 걸려 있는 포트에서 통신하려면 nc가 아니라 openssl s_client를 사용. ncat을 이용할 수도 있음.

password: kS0Hf0u5HiXFwKMKFqXvPdOTNGGa0X8V

```
bandit15@bandit:~$ openssl s_client -connect localhost:30001
Connecting to 127.0.0.1
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 CN=SnakeOil
verify error:num=18:self-signed certificate
verify return:1
depth=0 CN=SnakeOil
verify return:1
---
Certificate chain
 0 s:CN=SnakeOil
   i:CN=SnakeOil
   a:PKEY: RSA, 4096 (bit); sigalg: sha256WithRSAEncryption
   v:NotBefore: Jun 10 03:59:50 2024 GMT; NotAfter: Jun  8 03:59:50 2034 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIFBzCCAu+gAwIBAgIUBLz7DBxA0IfojaL/WaJzE6Sbz7cwDQYJKoZIhvcNAQEL
BQAwEzERMA8GA1UEAwwIU25ha2VPaWwwHhcNMjQwNjEwMDM1OTUwWhcNMzQwNjA4
MDM1OTUwWjATMREwDwYDVQQDDAhTbmFrZU9pbDCCAiIwDQYJKoZIhvcNAQEBBQAD
ggIPADCCAgoCggIBANI+P5QXm9Bj21FIPsQqbqZRb5XmSZZJYaam7EIJ16Fxedf+
jXAv4d/FVqiEM4BuSNsNMeBMx2Gq0lAfN33h+RMTjRoMb8yBsZsC063MLfXCk4p+
```

```
read R BLOCK
pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7
Correct!
kS0Hf0u5HiXFwKMKFqXvPdOTNGGa0X8V
```

```
bandit15@bandit:~$ ncat --ssl localhost 30001
pbLYuZtTg4MgaqfJx8jbA9gKKGqM68A7
Correct!
kS0Hf0u5HiXFwKMKFqXvPdOTNGGa0X8V
```

Level 16 -> Level 17

solution: nmap -p 31000-32000 localhost ->nmap -p 31046, 31518, 31691, 31790, 31960 -sV localhost -> ncat –ssl localhost 31790

nmap은 네트워크 스캐닝 도구. -p 옵션은 포트 번호를 지정할 수 있음. -sV 옵션은 열린 포트의 정체 확인

password: pWXMAZoxGC8JmDMfmT5MGEsobMM3vnj2

```
bandit16@bandit:~$ nmap -p 31000-32000 localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 08:04 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00013s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 996 closed tcp ports (conn-refused)
PORT      STATE SERVICE
31046/tcp open  unknown
31518/tcp open  unknown
31691/tcp open  unknown
31790/tcp open  unknown
31960/tcp open  unknown

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
bandit16@bandit:~$ nmap -p 31046,31518,31691,31790,31960 -sV localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-14 08:07 +0000
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000051s latency).
Other addresses for localhost (not scanned): ::1

PORT      STATE SERVICE     VERSION
31046/tcp open  echo
31518/tcp open  ssl/echo
31691/tcp open  echo
31790/tcp open  ssl/unknown
31960/tcp open  echo
```

```
bandit16@bandit:~$ ncat --ssl localhost 31790
kS0Hf0u5HiXFwKMKFqXvPdOTNGGa0X8V
Correct!
-----BEGIN OPENSSH PRIVATE KEY-----
b3BlbnNzaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAABlwAAAAdzc2gtcn
NhAAAAAwEAAQAAAYEAvdSaw8j1FQ2DjtbQPGiEVtqEG5kt3g71uDlixg42vRN2MvWRVnGQ
t4k9T9tDWaisnn+6I4RCkhEzw231WA6KVc0Sd0+6/6Cp1Egp4o4l+xf5gPNo7A2OqjqN67
Hhy6I71GBjyUBnp6vEtkI3WZmZtuxpCMPyHSy7m56lipJFddKEOUCX21hNWWy2SAZQFBub
3M1hrcar5cA4pCFJ2AmjSsOP4yRbdERh3vZTGNjKe2x+ze4jf2/Y/uNdmixdaAMuD8to4Y
f7JylXL/+ohzasOYM0iNFvr8gkOOc11xuTNdbGNmu1Ff3Vp1qtJNB600EWrBt9H4xl7/WX
```

```
bandit17@bandit:~$ cd /etc/bandit_pass
bandit17@bandit:/etc/bandit_pass$ cat bandit17
pWXMAZoxGC8JmDMfmT5MGEsobMM3vnj2
```

Level 17 -> Level 18

solution: diff passwords.new passwords.old

diff는 두 파일을 줄 단위로 비교하여 차이점을 찾음. <는 첫번째 파일의 내용, >는 두번째 파일의 내용

password: OQxXZjELndr90zuhOTDYBEomI0SZITXI

```
bandit17@bandit:~$ ls
passwords.new  passwords.old
bandit17@bandit:~$ diff passwords.new passwords.old
42c42
< OQxXZjELndr90zuhOTDYBEomI0SZITXI
---
> icUh23IUytZLIYhcCaXL18agiSIqymBc
bandit17@bandit:~$ exit
```

Level 18 -> Level 19

solution: ssh -p 2220 bandit18@bandit.labs.overthewire.org ‘cat readme’

ssh 명령어 뒤에 따옴표를 사용하면 로그인이 성공했을 때 따옴표 안의 명령어를 실행하고 자동으로 로그아웃함.

password: KpsOfPkcP7i1FlIExk2QEjyt6dw8dxZI

```
hyuk:~ $ ssh -p 2220 bandit18@bandit.labs.overthewire.org 'cat readme'
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit18@bandit.labs.overthewire.org's password:
KpsOfPkcP7i1FlIExk2QEjyt6dw8dxZI
```

Level 19 -> Level 20

solution: ./bandit20-do cat /etc/bandit_pass/bandit20

setuid는 실행 파일이 실행되는 동안 파일 소유자의 권한으로 실행되도록 함.

password: 4pljcunZ0fK2vmp3lwfG8Vf7VhxD6pOA

```
bandit19@bandit:~$ ls
bandit20-do
bandit19@bandit:~$ ./bandit20-do
Run a command as another user.
  Example: ./bandit20-do whoami
bandit19@bandit:~$ ./bandit20-do cat /etc/bandit_pass/bandit20
4pIjcunZ0fK2vmp3IwfG8Vf7VhxD6pOA
```

Level 20 -> Level 21

solution: tmux new -s bandit -> ncat -l -p 33333 -> ./suconnect 33333, 이전 단계의 password 입력

tmux는 하나의 터미널 창에서 여러 개의 세션, 윈도우, 패널을 생성하고 관리할 수 있게 해주는 터미널 멀티플렉서. tmux new -S로 새로운 세션을 만듦. tmux a -t로 세션으로 진입. Ctrl + b를 누른 후 다른 키와 조합하여 단축키 사용. %는 좌/우 화면 분할, “는 상/하 화면 분할, 방향키는 분할된 칸 커서 이동, d는 세션 나가기, x는 현재 커서가 있는 화면 닫기.

password: bW9kBv5WC3P4yoDyf12LSdGuNz5ka6hY

```
bandit20@bandit:~$ ncat -l -p 33333
4pIjcunZ0fK2vmp3IwfG8Vf7VhxD6pOA
bW9kBv5WC3P4yoDyf12LSdGuNz5ka6hY
bandit20@bandit:~$
```

```
bandit20@bandit:~$ ls
suconnect
bandit20@bandit:~$ ./suconnect 33333
Read: 4pIjcunZ0fK2vmp3IwfG8Vf7VhxD6pOA
Password matches, sending next password
bandit20@bandit:~$
```

Level 21 -> Level 22

solution: cat cronjob_bandit22 -> cat /usr/bin/cronjob_bandit22.sh -> cat 임시파일

cron은 특정 시간, 날짜, 주기마다 스크립트나 명령어를 자동으로 실행해 주는 시간 기반 작업 스케줄러. 처음 * 5개는 분/시/일/월/요일 순서.

password: RYVux2rHEm9tiXHmLFzuR7Vhx6AZQMEz

```
bandit21@bandit:/etc/cron.d$ ls
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_cleanup    otw-tmp-dir
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_resetpw_job
bandit21@bandit:/etc/cron.d$ cat cronjob_bandit22
@reboot bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
* * * * * bandit22 /usr/bin/cronjob_bandit22.sh &> /dev/null
bandit21@bandit:/etc/cron.d$ cat /usr/bin/cronjob_bandit22.sh
#!/bin/bash
chmod 644 /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
cat /etc/bandit_pass/bandit22 > /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
bandit21@bandit:/etc/cron.d$ cat /tmp/t706lds9S0RqQh9aMcz6ShpAoZKF7fgv
RYVux2rHEm9tiXHmLFzuR7Vhx6AZQMEz
```

Level 22 -> Level 23

solution: echo “I am User bandit23” | md5sum | cut -d ‘ ‘ -f 1

password: gKXDTAXnIz3OBxiPjRZ2uqutUlPZrBsw

```
bandit22@bandit:~$ cd /etc/cron.d
bandit22@bandit:/etc/cron.d$ ls
behemoth4_cleanup  cronjob_bandit22  cronjob_bandit24  leviathan5_cleanup    otw-tmp-dir
clean_tmp          cronjob_bandit23  e2scrub_all       manpage3_resetpw_job
bandit22@bandit:/etc/cron.d$ cat cronjob_bandit23
@reboot bandit23 /usr/bin/cronjob_bandit23.sh  &> /dev/null
* * * * * bandit23 /usr/bin/cronjob_bandit23.sh  &> /dev/null
bandit22@bandit:/etc/cron.d$ cat /usr/bin/cronjob_bandit23.sh
#!/bin/bash

myname=$(whoami)
mytarget=$(echo I am user $myname | md5sum | cut -d ' ' -f 1)

echo "Copying passwordfile /etc/bandit_pass/$myname to /tmp/$mytarget"

cat /etc/bandit_pass/$myname > /tmp/$mytarget
bandit22@bandit:/etc/cron.d$ echo "I am user bandit23" | md5sum | cut -d ' ' -f 1
8ca319486bfbbc3663ea0fbe81326349
bandit22@bandit:/etc/cron.d$ cat /tmp/8ca319486bfbbc3663ea0fbe81326349
gKXDTAXnIz3OBxiPjRZ2uqutUlPZrBsw
```

Level 23 -> Level 24

solution: cat /usr/bin/cronjob_bandit24.sh -> 쉘스크립트를 작성하여 다른 사용자에게 실행 권한을 부여하고 bandit24의 권한으로 password를 읽어서 파일 생성

쉘스크립트의 첫 줄에는 #!/bin/bash를 적어줘야 함. cronjob_bandit24를 보면 매분마다 쉘스크립트를 실행하는데 쉘스크립트는 특정 디렉터리의 파일을 실행한 뒤 삭제. 따라서 임시 디렉터리에서 쉘스크립트를 작성 후 해당 디렉터리에 복사. cronjob_bandit24의 실행 주체는 bandit24이므로 /usr/bin/cronjob_bandit24.sh가 실행될 때 whoami 명령어의 결과는 bandit24가 됨.

임시 디렉터리를 만들면 기본적으로 다른 사용자에게 권한이 부여되지 않은 상태여서 /usr/bin/cronjob_bandit24.sh가 실행되어도 내가 만든 쉘스크립트에 따라 임시 디렉터리에 파일이 생성되지 않음. 따라서 임시 디렉터리에도 권한을 설정해야 함.

password: hVQMk3lJNsmQ7VF3ubyrNNBom7BOgVXv

```
bandit23@bandit:/etc/cron.d$ cat cronjob_bandit24
@reboot bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
* * * * * bandit24 /usr/bin/cronjob_bandit24.sh &> /dev/null
bandit23@bandit:/etc/cron.d$ cat /usr/bin/cronjob_bandit24.sh
#!/bin/bash

shopt -s nullglob

myname=$(whoami)

cd /var/spool/"$myname"/foo || exit
echo "Executing and deleting all scripts in /var/spool/$myname/foo:"
for i in * .*;
do
    if [ "$i" != "." ] && [ "$i" != ".." ];
    then
        echo "Handling $i"
        owner="$(stat --format "%U" "./$i")"
        if [ "${owner}" = "bandit23" ] && [ -f "$i" ]; then
            timeout -s 9 60 "./$i"
        fi
        rm -rf "./$i"
    fi
donebandit23@bandit:/etc/cron.d$
```

```
bandit23@bandit:/etc/cron.d$ mktemp -d
/tmp/tmp.ODytdpQMu2
bandit23@bandit:/etc/cron.d$ cd /tmp/tmp.ODytdpQMu2
bandit23@bandit:/tmp/tmp.ODytdpQMu2$ vi key.sh
bandit23@bandit:/tmp/tmp.ODytdpQMu2$ cp key.sh /var/spool/bandit24/foo
```

```
#!/bin/bash

cat /etc/bandit_pass/bandit24 > /tmp/tmp.ODytdpQMu2/bandit24_password
```

```
bandit23@bandit:/tmp/tmp.ODytdpQMu2$ chmod 777 /tmp/tmp.ODytdpQMu2
bandit23@bandit:/tmp/tmp.ODytdpQMu2$ chmod 777 key.sh
bandit23@bandit:/tmp/tmp.ODytdpQMu2$ cp key.sh /var/spool/bandit24/foo
bandit23@bandit:/tmp/tmp.ODytdpQMu2$ ls
bandit24_password  key.sh
bandit23@bandit:/tmp/tmp.ODytdpQMu2$ cat bandit24_password
hVQMk3lJNsmQ7VF3ubyrNNBom7BOgVXv
```

Level 24 -> Level 25

solution: ./스크립트파일 | ncat localhost 30002

0부터 9999까지의 숫자를 무작위로 대입하여 전송해야 하므로 쉘스크립트를 작성. {1..9999}는 1부터 9999까지. 파이썬의 range와 비슷하지만 끝을 포함. 반복문의 시작과 끝을 do ~ done으로 나타냄. grep의 -v 옵션을 이용하면 특정 단어를 포함하지 않는 줄만

찾을 수 있음.

password: SoHfqMOEqIX2IYKVciZxvgpR9a2Djx4P

```
bandit24@bandit:/tmp/tmp.F36pNCAvV7$ ncat localhost 30002
I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a sing
le line, separated by a space.
```

```bash
#!/bin/bash

bandit24="hVQMk3lJNsmQ7VF3ubyrNNBom7BOgVXv"

for i in {1..9999}
do
        echo "$bandit24 $i"
done
```

```
bandit24@bandit:/tmp/tmp.F36pNCAvV7$ ./key | ncat localhost 30002
I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a sing
le line, separated by a space.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
```

```
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Wrong! Please enter the correct current password and pincode. Try again.
Correct!
The password of user bandit25 is SoHfqMOEqIX2IYKVciZxvgpR9a2Djx4P
```

```
bandit24@bandit:/tmp/tmp.F36pNCAvV7$ ./key | ncat localhost 30002 | grep -v "Wrong"
I am the pincode checker for user bandit25. Please enter the password for user bandit24 and the secret pincode on a sing
le line, separated by a space.
Correct!
The password of user bandit25 is SoHfqMOEqIX2IYKVciZxvgpR9a2Djx4P
```

Level 25 -> Level 26

solution: 창 줄여서 실행 후 more의 대기 상태에서 v 입력 -> :set shell /bin/bash -> :shell -> cat /etc/bandit_pass/bandit26

password: jHdv2ELQhT22BkprMNDjybZDAkw1zeBJ

badnit26.sshkey를 이용해서 bandit26에 로그인하니 연결하자마자 끊김. /etc/passwd 파일에서 다른 단계들은 /bin/bash를 실행하는데 반해 bandit26은 /usr/bin/showtext를 실행함. 파일을 확인해보니 exec more ~/text.txt 명령이 존재. exec는 현재 실행 중인 프로그램인 showtext를 종료하고 메모리에서 삭제한 후 그 자리를 more 프로그램이 대체하도록 하는 명령어. more는 텍스트 파일 내용을 화면 크기에 맞춰서 출력하는 프로그램. 따라서 bandit26에 로그인할 때 창 크기를 줄여서 로그인하면 more가 텍스트 전체를 한 번에 출력하지 못하므로 대기 상태가 됨. more는 실행 중에 v를 누르면 vi 편집기로 들어갈

수 있음. 명령창에서 쉘을 bash로 바꾸고 /bin/bash를 실행하는 자식 프로세스를 생성하여 password를 확인.

```
hyuk:~ $ ssh -p 2220 -i bandit26.sshkey bandit26@bandit.labs.overthewire.org
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
```

```
   For support, questions or comments, contact us on discord or IRC.

   Enjoy your stay!

  _                     _ _ _   ___   __
 | |                   | (_) | |__ \ / /
 | |__   __ _ _ __   __| |_| |_   ) / /_
 | '_ \ / _` | '_ \ / _` | | __| / / '_ \
 | |_) | (_| | | | | (_| | | |_ / /| (_) |
 |_.__/ \__,_|_| |_|\__,_|_|\__|____\___/
Connection to bandit.labs.overthewire.org closed.
```

```
bandit21:x:11021:11021:bandit level 21:/home/bandit21:/bin/bash
bandit22:x:11022:11022:bandit level 22:/home/bandit22:/bin/bash
bandit23:x:11023:11023:bandit level 23:/home/bandit23:/bin/bash
bandit24:x:11024:11024:bandit level 24:/home/bandit24:/bin/bash
bandit25:x:11025:11025:bandit level 25:/home/bandit25:/bin/bash
bandit26:x:11026:11026:bandit level 26:/home/bandit26:/usr/bin/showtext
bandit27:x:11027:11027:bandit level 27:/home/bandit27:/bin/bash
bandit28:x:11028:11028:bandit level 28:/home/bandit28:/bin/bash
```

```
bandit25@bandit:/etc$ cat /usr/bin/showtext
#!/bin/sh

export TERM=linux

exec more ~/text.txt
exit 0
```

Level 26 -> Level 27

solution: ./bandit27-do cat /etc/bandit_pass/bandit27

password: STJLJBRRphMxKB392CT4iOr5CbzPU9ER

```
bandit26@bandit:~$ ls
bandit27-do  text.txt
bandit26@bandit:~$ ./bandit27-do cat /etc/bandit_pass/bandit27
STJLJBRRphMxKB392CT4iOr5CbzPU9ER
```

Level 27 -> Level 28

solution: git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-git/repo -> cd repo -> cat README

password: y8Yd2ssKcpHpud7UvOSOxwamRMzIGIeQ

```
hyuk:~ $ git clone ssh://bandit27-git@bandit.labs.overthewire.org:2220/home/bandit27-git/repo
Cloning into 'repo'...
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit27-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
hyuk:~ $ ls
SSH  repo
hyuk:~ $ cd repo
hyuk:~/repo $ ls
README
hyuk:~/repo $ cat README
The password to the next level is: y8Yd2ssKcpHpud7UvOSOxwamRMzIGIeQ
```

Level 28 -> Level 29

solution: git log -> git show [커밋 해시값]

git log를 이용해서 커밋 기록을 볼 수 있으며 git show를 이용해서 커밋의 내용을 확인할 수 있음.

password: Em7eGtqaMySwNFjCpwzzHhLhospOcdt0

```
hyuk:~ $ git clone ssh://bandit28-git@bandit.labs.overthewire.org:2220/home/bandit28-git/repo
Cloning into 'repo'...
                         _                     _ _ _
                        | |__   __ _ _ __   __| (_) |_
                        | '_ \ / _` | '_ \ / _` | | __|
                        | |_) | (_| | | | | (_| | | |_
                        |_.__/ \__,_|_| |_|\__,_|_|\__|


                      This is an OverTheWire game server.
            More information on http://www.overthewire.org/wargames

backend: gibson-0
bandit28-git@bandit.labs.overthewire.org's password:
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), 800 bytes | 114.00 KiB/s, done.
Resolving deltas: 100% (2/2), done.
hyuk:~ $ ls
SSH  repo
```

```
hyuk:~/repo $ cat README.md
# Bandit Notes
Some notes for level29 of bandit.

## credentials

- username: bandit29
- password: xxxxxxxxxx

hyuk:~/repo $ git log
commit e2e1de5396037bafb23e9bb37c12ebea9b911cfd (HEAD -> master, origin/master, origin/HEAD)
Author: Morla Porla <morla@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    fix info leak

commit 2678cfadd8f2a347bc23e1ea491f702e5b184709
Author: Morla Porla <morla@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    add missing data

commit 9530d526c22b9e6e6ae11070ef8ff8ee21eb2e02
Author: Ben Dover <noone@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    initial commit of README.md
```

```
hyuk:~/repo $ git show e2e1de5396037bafb23e9bb37c12ebea9b911cfd
commit e2e1de5396037bafb23e9bb37c12ebea9b911cfd (HEAD -> master, origin/master, origin/HEAD)
Author: Morla Porla <morla@overthewire.org>
Date:   Wed Jun 24 14:59:20 2026 +0000

    fix info leak

diff --git a/README.md b/README.md
index 42331d9..5c6457b 100644
--- a/README.md
+++ b/README.md
@@ -4,5 +4,5 @@ Some notes for level29 of bandit.
 ## credentials

 - username: bandit29
-- password: Em7eGtqaMySwNFjCpwzzHhLhospOcdt0
+- password: xxxxxxxxxx
```

Level 29 -> Level 30

solution: git checkout dev -> git show dev

git log —-graph —-oneline —-all은 커밋 트리를 보여줌. –graph는 커밋 트리, --oneline은 한 줄 요약, --all은 현재 위치하는 브랜치뿐만 아니라 다른 브랜치와 원격브랜치까지 보여줌.

password: jq9Dfg2rXsfYsWMgFuKlXhphjdH7USgX

```
hyuk:~/repo $ git log --graph --oneline --all
* 0bf8160 (origin/dev, dev) add data needed for development
* 1b95ced add gif2ascii
| * a0e9e9c (origin/sploits-dev) add some silly exploit, just for shit and giggles
|/
* b607fba (HEAD -> master, origin/master, origin/HEAD) fix username
* 84c16f8 initial commit of README.md
hyuk:~/repo $ git show dev
commit 0bf8160435c51882945ce56ae74a2798dd3c57f3 (origin/dev, dev)
Author: Morla Porla <morla@overthewire.org>
Date:   Wed Jun 24 14:59:22 2026 +0000

    add data needed for development

diff --git a/README.md b/README.md
index 1af21d3..d395d04 100644
--- a/README.md
+++ b/README.md
@@ -4,5 +4,5 @@ Some notes for bandit30 of bandit.
 ## credentials

 - username: bandit30
-- password: <no passwords in production!>
+- password: jq9Dfg2rXsfYsWMgFuKlXhphjdH7USgX
```

Level 30 -> Level 31

solution: git tag -> git show secret

태그를 커밋에 붙이지 않고 BLOB(Binary Large Object)에 직접 붙일 수도 있음. blob은 파일 내용을 저장하는 깃의 객체 유형.

password: 82NkymblpGBYmIXG6ZQ8YldBYstHpfUf

```
hyuk:~/repo $ git log --graph --oneline --all
* 929c564 (HEAD -> master, origin/master, origin/HEAD) initial commit of README.md
hyuk:~/repo $ git tag
secret
hyuk:~/repo $ git show secret
82NkymblpGBYmIXG6ZQ8YldBYstHpfUf
```

Level 31 -> Level 32

solution: echo “May I come in?” > key.txt -> git add -f key.txt -> git commit -m “~” -> git push

git add의 -f 옵션을 사용하면 .gitignore 파일에 의해 제한된 파일도 add 할 수 있음.

password: pWuj5jBQ6IgV0NXwiH6g1pXRF8S1YvbT

```
hyuk:~/repo $ ls
README.md
hyuk:~/repo $ cat README.md
This time your task is to push a file to the remote repository.

Details:
    File name: key.txt
    Content: 'May I come in?'
    Branch: master

hyuk:~/repo $ echo "May I come in?" > key.txt
hyuk:~/repo $ ls
README.md  key.txt
hyuk:~/repo $ git add key.txt
The following paths are ignored by one of your .gitignore files:
key.txt
hint: Use -f if you really want to add them.
hint: Turn this message off by running
hint: "git config advice.addIgnoredFile false"
hyuk:~/repo $ git add -f key.txt
hyuk:~/repo $ git commit -m "create key.txt"
[master aaa1a9d] create key.txt
 1 file changed, 1 insertion(+)
 create mode 100644 key.txt
hyuk:~/repo $ git push
```

```
backend: gibson-0
bandit31-git@bandit.labs.overthewire.org's password:
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 322 bytes | 322.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote: ### Attempting to validate files... ####
remote:
remote: .oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.oOo.
remote:
remote: Well done! Here is the password for the next level:
remote: pWuj5jBQ6IgV0NXwiH6g1pXRF8S1YvbT
```

Level 32 -> Level 33

solution: $0 -> cat /etc/bandit_pass/bandit33

$0은 현재 실행 중인 프로그램의 이름을 저장하는 변수. uppershell이 실행 중인 상태에서 $0을 입력하면 uppershell은 /bin/sh로 $0을 넘겨서 실행을 요청. /bin/sh 입장에서 $0은 /bin/sh이므로 새로운 /bin/sh 프로세스를 띄움. /bin/sh의 기본 프롬프트는 $.

password: u4P2CyPOwPGLe94RdD9Uo2FxFwvnFswM

```
WELCOME TO THE UPPERCASE SHELL
>> ls
sh: 1: LS: Permission denied
>> $0
$ bash
bandit33@bandit:~$ cat /etc/bandit_pass/bandit33
u4P2CyPOwPGLe94RdD9Uo2FxFwvnFswM
bandit33@bandit:~$ ls
uppershell
```